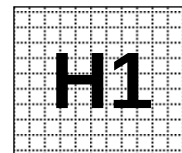


Civics Group	Index Number	Name (use BLOCK LETTERS)
--------------	--------------	--------------------------



**ST. ANDREW'S JUNIOR COLLEGE
2014 Preliminary Examination**

H1 BIOLOGY

8875/2

Paper 2: Core

Tuesday

2 September 2014

2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Compulsory question to be answered on writing paper provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/9½
2	/8
3	/10
4	/12½
5	/20
Total	/60

This document consists of **16** printed pages.

[Turn over

Section A

Answer **all** questions.

QUESTION 1

The fruitfly, *Drosophila*, has many different species. Three of these species, *Drosophila pseudoobscura*, *D. persimilis* and *D. miranda*, are thought to be closely related.

Samples of these three species were collected from the western United States of America.

Fig. 1.1 shows where these species naturally occur.

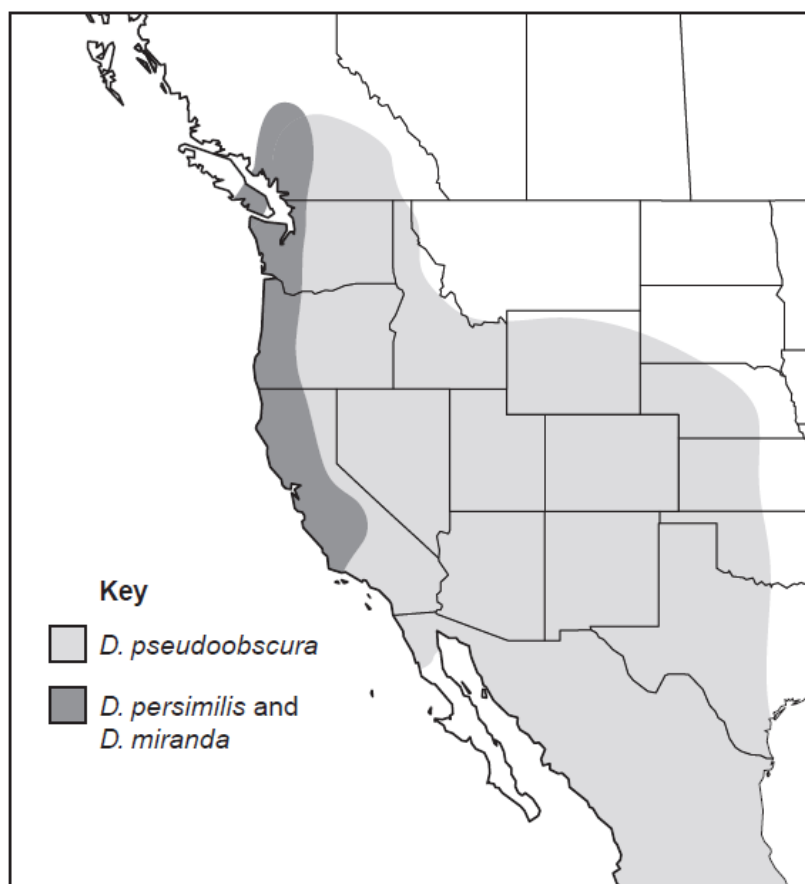


Fig. 1.1

The base sequences of four regions of DNA of each species were sequenced. The divergence of these base sequences in *D. pseudoobscura* and *D. persimilis* from the sequences in *D. miranda* was calculated. The results are shown in Table 1.2.

DNA region	<i>Drosophila</i> species	percentage divergence of base sequence from that of <i>D. miranda</i>
1	<i>pseudoobscura</i>	2.5
	<i>persimilis</i>	2.4
2	<i>pseudoobscura</i>	8.1
	<i>persimilis</i>	7.3
3	<i>pseudoobscura</i>	2.1
	<i>persimilis</i>	1.7
4	<i>pseudoobscura</i>	1.9
	<i>persimilis</i>	1.7

Table 1.2

- (a) With reference to Table 1.2, describe the evidence that *D.miranda* may be more closely related to *D.persimilis* than to *D.pseudoobscura*.

.....

.....

.....

.....

.....[1½]

- (b) Suggest why there is more divergence in some regions of DNA than in others.

.....

.....

.....

.....[1]

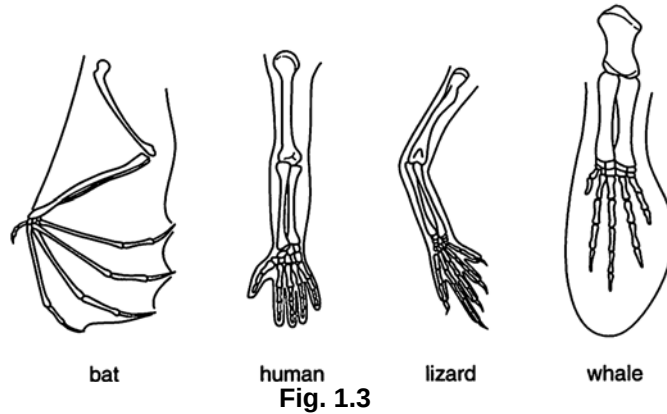
- (c) The area where *D. pseudoobscura* is found is separated from the areas where the other two species are found by a high range of mountains.

Explain how the species *D. pseudoobscura* could have evolved from a population of *D. miranda*.

[4]

Besides the use of DNA sequences to analyse evolutionary relationships between species, anatomical homologies are also used.

- (d) Explain how anatomical homologies observed in the forelimbs of the animals shown in Fig. 1.3 support Darwin's theory of natural selection.



.....

.....

.....

.....

.....

.....

.....

.....[3]

[Total : 9½]

(b) Suggest why some cells did not stained blue at 20⁰C.

.....
.....[1]

(c) Suggest one change that occurred to the cell surface membranes of the yeast cells at temperatures above 60⁰C.

.....[½]

(d) Explain why the stained yeast cells lost their colour at higher temperatures.

.....
.....
.....
.....
.....[1½]

The student placed a small sample of the yeast suspension on a microscope slide and observed it under high power.

Fig.2.2 shows what the student observed.

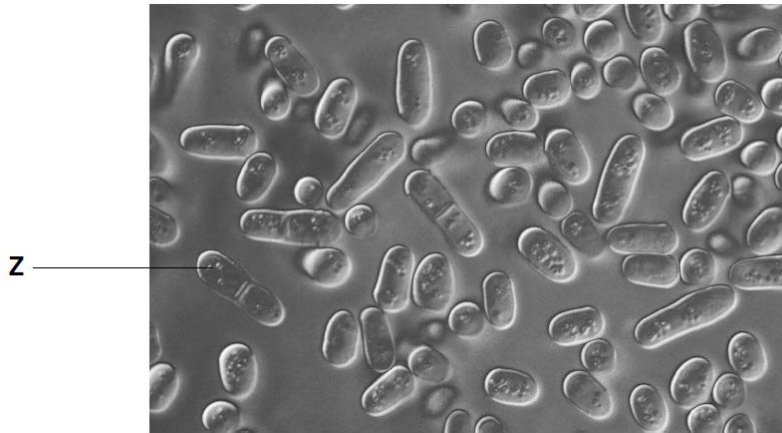


Fig. 2.2

Cell Z is undergoing a process known as budding. Budding is a form of asexual reproduction.

(e) Explain the importance of mitosis in budding.

.....

.....

.....

.....

.....[1½]

[Total : 8]

QUESTION 3

Fig.3.1 shows the relationship between various metabolic processes.

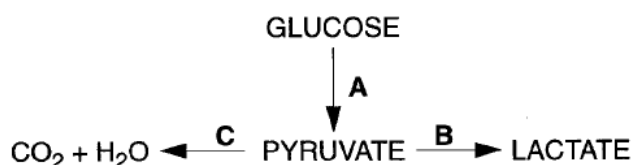


Fig. 3.1

(a) (i) Identify the three metabolic processes. [1½]

A

B

C

(ii) State the letter of the pathway in which acetyl coenzyme A is involved.

.....[½]

(iii) State the letter of the pathway in which ATP is utilized.

.....[½]

In an investigation, mammalian liver cells were homogenized and the resulting homogenate centrifuged. Portions containing only nuclei, ribosomes, mitochondria and cytosol were each isolated. Samples of each portion, and of the complete homogenate, were incubated in four ways,

- 1 with glucose
- 2 with pyruvate
- 3 with glucose plus cyanide
- 4 with pyruvate plus cyanide

Cyanide inhibits carriers in the electron transport chain, such as cytochromes. After incubation, the presence or absence of carbon dioxide and lactate in each sample were determined.

The results are summarised in Table 3.2.

	samples of homogenate									
	complete		nuclei only		ribosomes only		mitochondria only		cytosol	
	carbon dioxide	lactate	carbon dioxide	lactate	carbon dioxide	lactate	carbon dioxide	lactate	carbon dioxide	lactate
1. glucose	✓	✓	X	X	X	X	X	X	X	✓
2. pyruvate	✓	✓	X	X	X	X	✓	X	X	✓
3. glucose and cyanide	X	✓	X	X	X	X	X	X	X	✓
4. pyruvate and cyanide	X	✓	X	X	X	X	X	X	X	✓

X = absent ✓ = present

Table. 3.2

- (b) (i) With reference to this investigation, name two organelles not involved in respiration.

.....[1]

- (ii) Explain why carbon dioxide is produced when mitochondria are incubated with pyruvate but not when incubated with glucose.

.....

.....

.....

.....

.....

.....

.....

.....[2½]

(iii) Explain why in the presence of cyanide, lactate production does occur, but not carbon dioxide production.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[3]

This investigation may be repeated using yeast cells instead of liver cells.

(c) State the products that would be formed by the incubation of glucose with cytosol from yeast.

.....[1]

[Total : 10]

QUESTION 4

- (a) In the fruit fly shown in Fig.4.1, *Drosophila melanogaster*, the gene for eye colour is sex-linked. The allele for red eye (R) is dominant to the allele for white eye (r). A student carried out two crosses.



Fig.4.1

The first cross involved crossing a red-eyed female with a white-eyed male. Two fruit flies of the F1 generation were then crossed. The following F2 phenotypes were produced 95 red-eyed females, 49 red-eyed males, 56 white-eyed males.

In the second cross, the student carried out a reciprocal cross.

- (i) Draw a genetic diagram in the space below to show the expected F1 and F2 generations from the reciprocal cross. You may also use the blank space on page 13 to complete your answer.

.....[4½]

- (b) The giant panda is an endangered species native to South Central China. Due to its conservation status, a lot of research studies have revolved around the giant panda. Recent research on a protein from the giant panda was carried out in 2008. This protein, called crystallin, is found in the lens of the eye, and has a sequence that has been highly conserved in mammals.

The steps in the procedure used in the study are summarized in Fig.4.2.

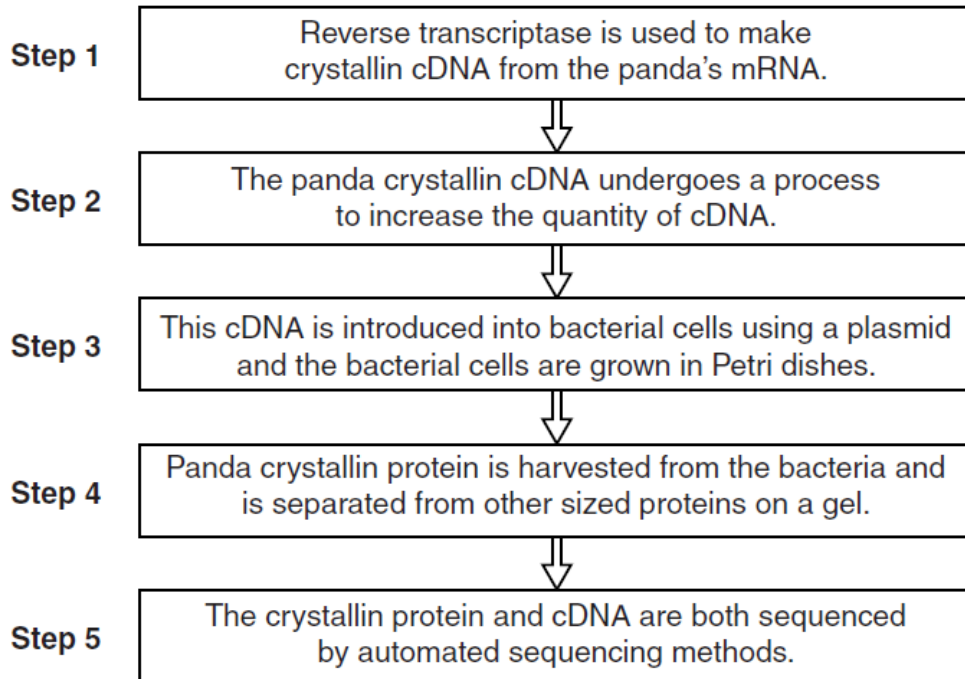


Fig.4.2

- (i) Identify the technique used in each case to carry out steps 2 and 4. [1]

Step 2

Step 4

- (ii) Explain why lens cells were used for the isolation of the giant panda's mRNA.

.....

[1]

- (iii) The giant panda crystallin protein cloned was 175 amino acids long, corresponding to a 528 base pair cDNA gene. Explain why a protein that is 175 amino acids long is coded for by 528 base pairs of DNA.

.....

[1½]

- (iv) In step 3 of the procedure shown in Fig.4.2, a plasmid was used in the cloning of the crystalline gene.

On the diagram of the plasmid shown in Fig.4.3, draw two named genetic markers and the other components that are required for the expression vector to be suitable for use in the cloning of the crystalline gene. You are to label these components with specific names where applicable. [3]

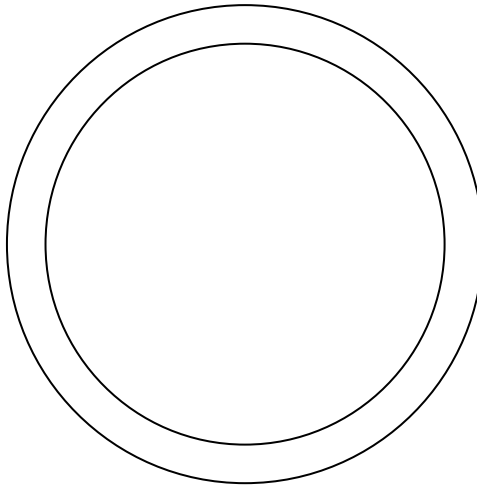


Fig.4.3

- (v) Explain how the plasmid in Fig.4.3 was introduced into bacterial cells in step 3 of the procedure.

.....

.....

.....

.....[1½]

[Total : 12½]

Section B

Answer the question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 **(a)** With reference to sickle cell anemia, explain how a change in the sequence of the DNA nucleotide affects the phenotype of the organism. [7]
- (b)** Describe the development of cancer as a multistep process. [8]
- The time course of an amylase digestion can be followed by measuring
- (c)** the rates of formation of the product, glucose.
- Describe how the concentration of glucose produced can be measured using a colorimeter. [5]

- **END OF PAPER** -